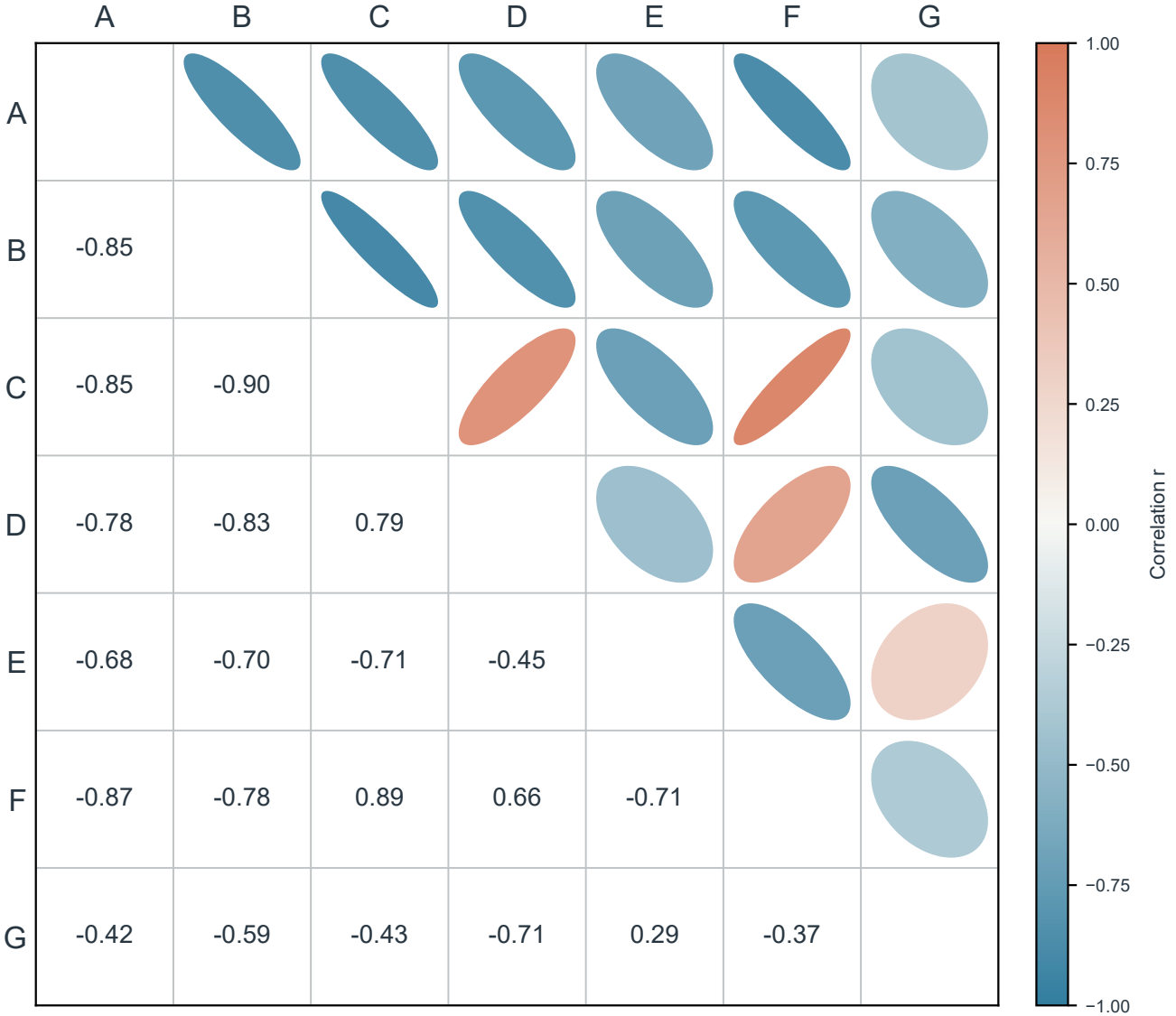Signed correlations with a neutral zero